

# ML-Based CO<sub>2</sub> Emissions Prediction and Bonus-Malus Web App

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## Abstract

Reducing vehicular CO<sub>2</sub> emissions is vital for addressing climate change, particularly given the transportation sector's significant contribution to greenhouse gas emissions. This project develops a data-driven machine learning framework to predict CO<sub>2</sub> emissions of passenger vehicles in France using detailed 2014 homologation data from UTAC and ADEME. Multiple models—including Lasso, Ridge, Random Forest, and XGBoost—are trained and evaluated to accurately estimate emissions based on vehicle technical specifications. Furthermore, the predictions are integrated with the 2014 French ecological bonus-malus policy, which imposes financial incentives or penalties according to emission levels, allowing quantification of the associated economic impacts. An interactive web application built with Dash enables users to input vehicle features, obtain real-time CO<sub>2</sub> emission predictions, and visualize corresponding bonus or malus values. This tool offers a practical solution to support policymakers, manufacturers, and consumers in understanding emission profiles and promoting environmentally informed decisions.

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The full codebase, data preprocessing steps, and web application are available on GitHub for replication: [https://github.com/cynthia-francis/Prediction\\_CO2\\_Emissions\\_Cars](https://github.com/cynthia-francis/Prediction_CO2_Emissions_Cars)

## Contents

|          |                              |          |
|----------|------------------------------|----------|
| <b>1</b> | <b>Introduction</b>          | <b>2</b> |
| <b>2</b> | <b>Literature Review</b>     | <b>3</b> |
| <b>3</b> | <b>Data</b>                  | <b>4</b> |
| 3.1      | Data Source . . . . .        | 4        |
| 3.2      | Data Exploration . . . . .   | 4        |
| 3.2.1    | Numerical Features . . . . . | 6        |

|          |                                                                                                      |           |
|----------|------------------------------------------------------------------------------------------------------|-----------|
| 3.2.2    | Categorical Features . . . . .                                                                       | 11        |
| <b>4</b> | <b>Machine Learning Models</b>                                                                       | <b>14</b> |
| 4.1      | Lasso Regression . . . . .                                                                           | 14        |
| 4.1.1    | Results . . . . .                                                                                    | 14        |
| 4.1.2    | Feature Importance . . . . .                                                                         | 14        |
| 4.2      | Ridge Regression . . . . .                                                                           | 15        |
| 4.2.1    | Results . . . . .                                                                                    | 16        |
| 4.2.2    | Feature Importance . . . . .                                                                         | 16        |
| 4.3      | Random Forest . . . . .                                                                              | 17        |
| 4.3.1    | Results . . . . .                                                                                    | 17        |
| 4.3.2    | Feature Importance . . . . .                                                                         | 17        |
| 4.4      | XGBoost . . . . .                                                                                    | 17        |
| 4.4.1    | Results . . . . .                                                                                    | 18        |
| 4.4.2    | Feature Importance . . . . .                                                                         | 18        |
| <b>5</b> | <b>Comparison Between Manufacturer Advertising in Auto Plus and Model-Derived Feature Importance</b> | <b>19</b> |
| 5.1      | Introduction to Auto Plus and 2014 Advertising Context . . . . .                                     | 19        |
| 5.2      | Model-Derived Feature Importance (Summary of Results) . . . . .                                      | 20        |
| 5.3      | Comparison . . . . .                                                                                 | 20        |
| <b>6</b> | <b>Web Application Architecture and Functionality</b>                                                | <b>20</b> |
| 6.1      | Data and Model Loading . . . . .                                                                     | 21        |
| 6.2      | Application Layout . . . . .                                                                         | 21        |
| 6.3      | Prediction Pipeline . . . . .                                                                        | 21        |
| 6.4      | User Experience Enhancements . . . . .                                                               | 22        |
| 6.5      | Deployment . . . . .                                                                                 | 22        |
| <b>7</b> | <b>Conclusion</b>                                                                                    | <b>22</b> |

## 1 Introduction

Climate change mitigation remains a critical global priority, with transportation identified as a significant contributor to greenhouse gas emissions ([Intergovernmental Panel on Climate Change \(IPCC\), 2022](#), [Creutzig et al., 2015](#)). In this context, understanding and predicting vehicular CO<sub>2</sub> emissions are essential for policymakers, manufacturers, and consumers aiming to reduce environmental impact. This project presents a data-driven approach to predicting CO<sub>2</sub> emissions from passenger vehicles in France using official homologation data from 2014. The dataset, provided by UTAC to ADEME, comprises detailed technical specifications of vehicles, including fuel type, hybrid status, vehicle mass, gearbox details, and real-world fuel consumption.

Leveraging this dataset, the project employs various machine learning models—such as Lasso regression, Ridge regression, Random Forest, and XGBoost—to estimate CO<sub>2</sub> emissions based on vehicle characteristics. Machine learning methods have proven effective for traffic and emissions prediction tasks ([Zhang et al., 2018](#), [Li et al., 2021](#)).

Beyond prediction, the project integrates these estimates with the 2014 French ecological bonus-malus policy, which imposes financial incentives or penalties on car owners depending on their vehicle’s emission levels (French Ministry for the Ecological Transition, 2014). This linkage allows quantification of the fiscal impact associated with different emission profiles (Simoes and Gouveia, 2017).

To facilitate user interaction, an interactive web application built with Dash enables real-time input of vehicle specifications, instantaneous CO<sub>2</sub> emissions prediction, and visualization of the corresponding bonus or malus. This tool serves as a practical platform for stakeholders to explore emission scenarios and their economic consequences, promoting informed decision-making aligned with environmental policies.

## 2 Literature Review

France’s bonus–malus feebate system and subsequent Electric Vehicle (EV) bonus reforms provide a rich policy background for studying how fiscal incentives shape new-car markets and aggregate emissions. Early evaluations of the 2008 bonus–malus find that the policy strongly shifted market shares toward bonus-eligible classes, but generated limited environmental gains and large fiscal costs. For example, D’Haultfoeuille et al. (2014) show that although sales of low-emission cars surged, many consumers clustered just below legislative thresholds, leading to modest reductions in average tailpipe emissions and substantial rebound effects that offset environmental benefits. Similarly, D’Haultfoeuille et al. (2011) compare ex ante predictions with realized outcomes and find that the observed shift toward bonus-eligible models was two to three times larger than standard price effects predicted, which explains why the policy’s intended budget neutrality failed in practice.

More recently, reforms tying subsidies to life-cycle environmental scores—such as the 2024 EV bonus reform—illustrate the policy’s evolution from simple emissions metrics to broader, supply-chain-aware criteria. Using a difference-in-differences design and a nested-logit demand model, Malgouyres et al. (2025) find that the reform successfully shifted purchases toward cleaner, domestically produced EVs, but also slowed overall EV adoption and temporarily increased tank-to-wheel emissions. Their simulations show that a budget-neutral redesign—maintaining the environmental score requirement but reallocating saved funds to eligible models—could preserve industrial benefits while restoring adoption rates and improving environmental outcomes.

Although this literature relies on rigorous causal and structural methods to evaluate policy effects and simulate counterfactuals, it largely focuses on post-implementation evaluation rather than on **predicting individual vehicle emissions from technical specifications**. This is the methodological gap that this project addresses. By training machine-learning models on 2014 vehicle data, I build a predictive tool that maps vehicle characteristics to expected CO<sub>2</sub> (g/km) emissions and automatically classifies the resulting emission into the 2014 French bonus/malus bands. This complements existing studies by enabling **ex ante forecasting**: given projected trends in vehicle technology or fleet composition, the tool can estimate

how the distribution of predicted emissions—and therefore bonus/malus eligibility shares—would evolve before a policy change is implemented. Such forecasts can help policymakers anticipate budgetary impacts and potential market responses without requiring full structural estimation.

## 3 Data

### 3.1 Data Source

The dataset used in this project is *[2014] Emissions de polluants, CO<sub>2</sub> et caractéristiques des véhicules commercialisés en France*, published by ADEME (Agence de la Transition Écologique) and publicly available on [data.gouv.fr](https://data.gouv.fr) under the [Etalab Open License](#). The data originate from the vehicle approval services of automobile manufacturers and are compiled by the Union Technique de l'Automobile, du Motocycle et du Cycle (UTAC), the body responsible for type-approval testing in France, in collaboration with the Ministry of Sustainable Development; they record measurements taken prior to market entry for each vehicle approved in 2014 and provide detailed technical and environmental attributes including fuel consumption (L/100 km or kWh/100 km for electric vehicles), CO<sub>2</sub> emissions (g/km), regulated air pollutant emissions under the Euro standards (e.g. NO<sub>x</sub>, particulate matter), and vehicle characteristics such as make and model, CNIT identifier, energy/fuel type (diesel, petrol, hybrid, electric), gearbox type and number of gears, vehicle category and body type; this comprehensive and well-structured dataset therefore supplies the technical and environmental indicators necessary to train and validate predictive models of CO<sub>2</sub> emissions and to analyse how vehicle attributes relate to emissions outcomes on the French market in 2014.

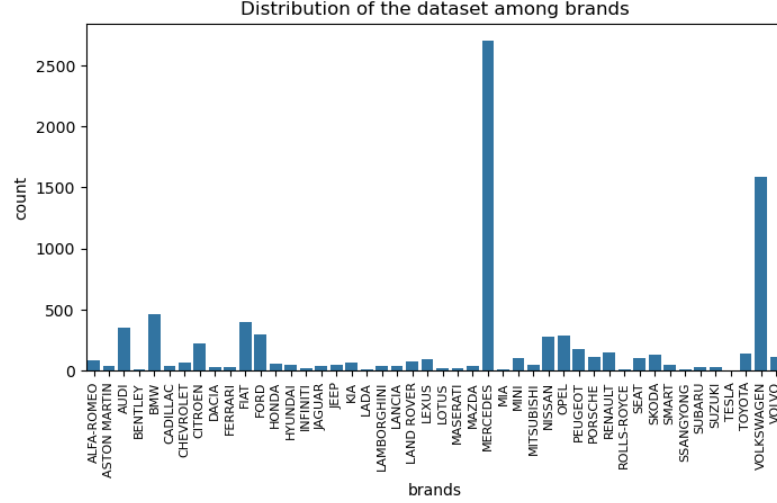
### 3.2 Data Exploration

Table 1 provides a glossary of the main variables in the dataset, grouped by category for clarity. These variables describe vehicle identification details, technical characteristics, fuel consumption, emissions, and other metadata.

| Vehicle Identification  |                                                       |
|-------------------------|-------------------------------------------------------|
| lib_mrqu_tac            | Brand (e.g., Renault, Peugeot)                        |
| lib_mod_doss            | Model name (e.g., Clio, 208)                          |
| lib_mod                 | Commercial model name                                 |
| dscom                   | Commercial description                                |
| cnit                    | CNIT code: official vehicle type identifier in France |
| tvv                     | Vehicle type/version code                             |
| cod_cbr                 | Fuel type (e.g., diesel, petrol)                      |
| gamme                   | Vehicle range/class (compact, SUV)                    |
| Carrosserie             | Body type (sedan, hatchback, etc.)                    |
| Power & Engine          |                                                       |
| hybride                 | Whether the car is a hybrid (O/N)                     |
| puiss_admin_98          | Administrative power (used for taxes)                 |
| puiss_max               | Maximum power (in kilowatts)                          |
| typ_boite_nb_rapp       | Gearbox type and number of gears                      |
| Fuel Consumption        |                                                       |
| conso_urb               | Urban fuel consumption (L/100 km)                     |
| conso_exurb             | Extra-urban fuel consumption (L/100 km)               |
| conso_mixte             | Combined (average) fuel consumption (L/100 km)        |
| Pollution & Emissions   |                                                       |
| co2                     | CO <sub>2</sub> emissions (g/km)                      |
| co_typ_1                | CO type I emissions (g/km)                            |
| hc                      | Hydrocarbon emissions (g/km)                          |
| nox                     | Nitrogen oxide emissions (g/km)                       |
| hcnnox                  | Combined HC + NO <sub>x</sub> emissions (g/km)        |
| ptcl                    | Particle emissions (g/km)                             |
| Weight                  |                                                       |
| masse_ordma_min         | Minimum curb weight (kg)                              |
| masse_ordma_max         | Maximum curb weight (kg)                              |
| Registration / Metadata |                                                       |
| champ_v9                | Field containing EURO emissions standard info         |
| date_maj                | Last update date                                      |

**Table 1:** Variable glossary for the 2014 vehicle dataset.

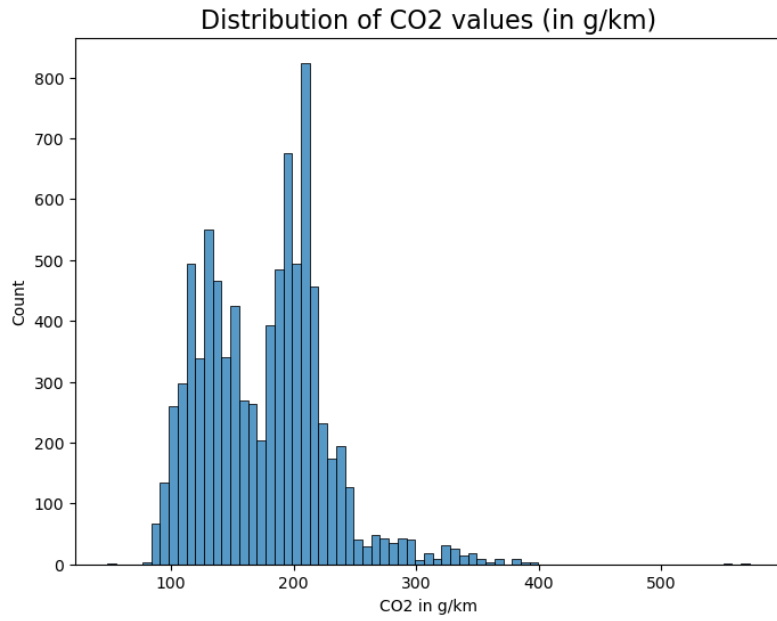
Initially, the dataset contained 55,044 entries. Upon inspection, several duplicate rows were identified and removed to ensure data quality and integrity. After this deduplication process, the dataset was reduced to 8,604 unique entries. According to the bar graph presented below, the brands Mercedes and Volkswagen remain the most prominently represented in the cleaned dataset.



**Fig. 1:** Brand distribution in the cleaned dataset showing Mercedes and Volkswagen as most prominent.

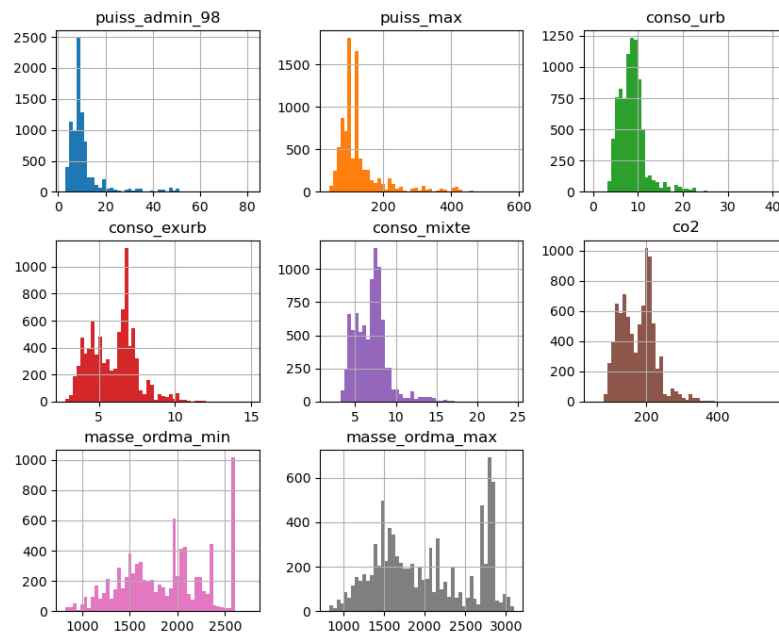
### 3.2.1 Numerical Features

To better understand the data, we first focus on exploring the numerical features, starting with our target variable: CO<sub>2</sub> emissions. Analyzing its distribution provides valuable insight into the types of vehicles represented in the dataset. As shown in Figure 2, the distribution of CO<sub>2</sub> emissions is clearly bimodal, indicating two main groups of vehicles — one with emissions centered around 140 g/km and another around 200 g/km. There are also a few noticeable outliers with much higher emissions.



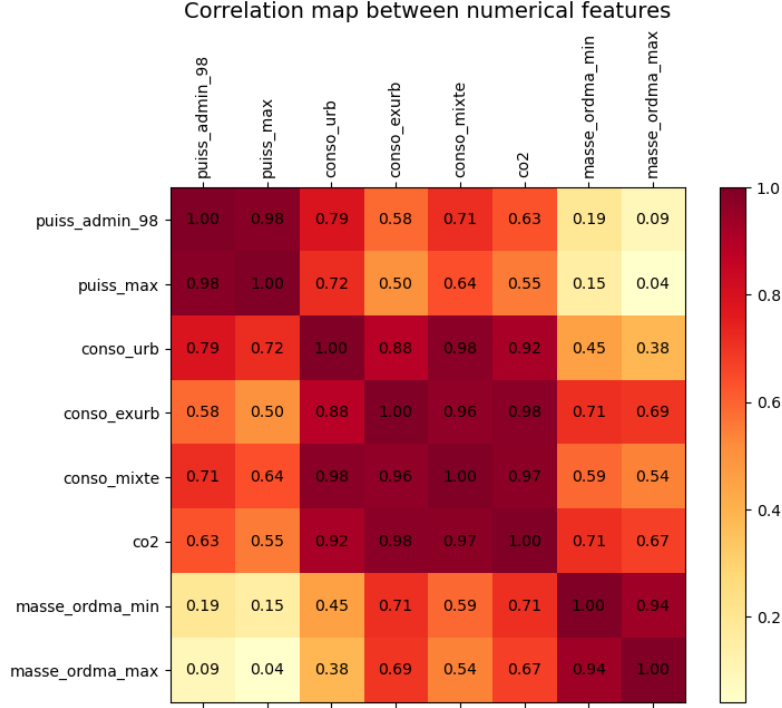
**Fig. 2:** Distribution of CO<sub>2</sub> emissions across vehicles.

Following this, we will explore the distributions of other numerical features and investigate their correlations with CO<sub>2</sub> emissions through a heatmap visualization.



**Fig. 3:** Distribution of numerical features across vehicles.



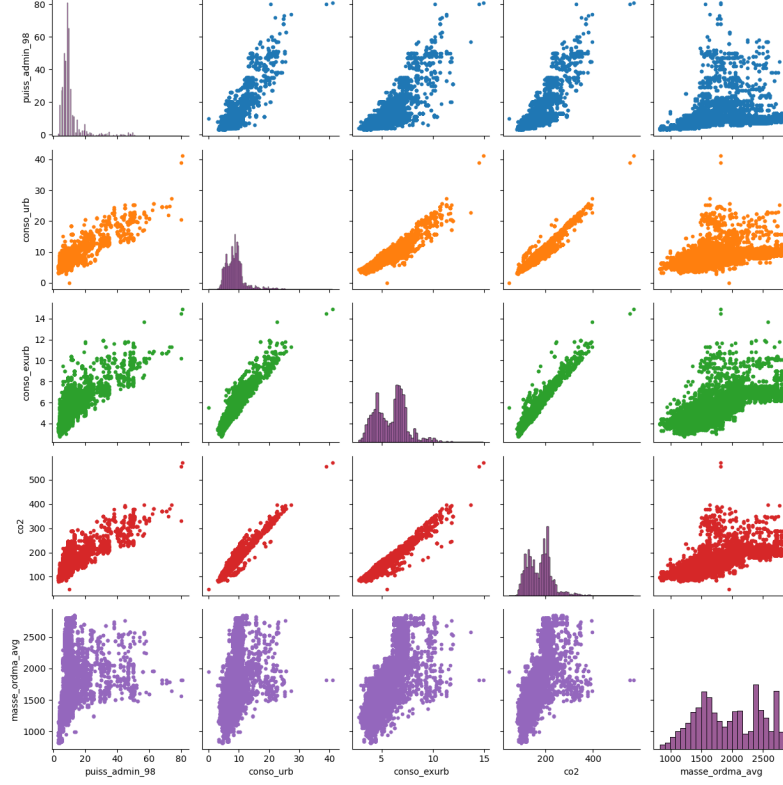


**Fig. 4:** Matrix displaying the correlations between numerical features.

The histograms reveal that most vehicles in the dataset are mid-sized, with moderate engine power and mass, and generally fall within average to good fuel efficiency ranges. Variables such as `puiss_admin_98` and `puiss_max` are right-skewed, indicating that while most vehicles have modest power ratings, a few high-performance outliers exist. Fuel consumption measures (`conso_urb`, `conso_exurb`, and `conso_mixte`) show relatively tight distributions centered around typical usage values, with fewer cases of extreme inefficiency. Vehicle mass variables (`masse_ordma_min` and `masse_ordma_max`) exhibit multiple peaks, suggesting distinct categories of vehicle classes (e.g., compact, mid-size, and heavier models).

When examining the correlation matrix, CO<sub>2</sub> emissions display strong positive correlations with all other features, reinforcing the potential for developing an accurate predictive model. Notably, certain features are almost perfectly correlated: `masse_ordma_min` and `masse_ordma_max` ( $r = 0.94$ ), `puiss_admin_98` and `puiss_max` ( $r = 0.98$ ), and `conso_mixte` with both `conso_urb` ( $r = 0.98$ ) and `conso_exurb` ( $r = 0.96$ ). These high correlations are consistent with the nature of the variables—administrative and maximum power are closely related measures, as are different fuel consumption contexts. Given this redundancy, it would be prudent to remove or combine some highly correlated variables to enhance interpretability and reduce multicollinearity in the modeling phase.

For instance, the dataset includes both minimum and maximum curb weights for each vehicle, reflecting the lightest and heaviest configurations of a given model. Our analysis reveals that in 50% of cases, the two values are identical, yet the average difference across the dataset is substantial ( $\sim 148$  kg), with some models exhibiting extreme variations of up to 833 kg. To preserve this meaningful variability while mitigating multicollinearity, we replace the original variables with their mean value, creating a single new feature, `masse_ordma_avg`, and removing the originals from the dataset.



**Fig. 5:** Pairplots displaying the relationships between numerical features.

After addressing these redundancies, we complement the correlation analysis with pairwise scatterplots to visualize the relationships between  $\text{CO}_2$  emissions and the most relevant numerical features (Figure 5). The plots reveal clear positive linear trends between  $\text{CO}_2$  and variables such as `puiss_admin_98`, `conso_urb`, `conso_exurb`, and `masse_ordma_avg`, confirming the strong associations observed in the correlation matrix. In particular, fuel consumption metrics (`conso_urb`, `conso_exurb`) display near-perfect alignment with  $\text{CO}_2$ , reflecting the direct impact of energy use on emissions. Power and weight, while slightly more dispersed, still exhibit

strong positive correlations, suggesting that heavier and more powerful vehicles tend to emit more CO<sub>2</sub>. These visualizations not only validate the statistical correlations but also provide an intuitive understanding of feature interactions, which is valuable for guiding feature selection in the modeling phase.

### 3.2.2 Categorical Features

In addition to the numerical variables, our dataset contains several categorical features that capture vehicle characteristics relevant to CO<sub>2</sub> emissions. Understanding these variables is essential for appropriate preprocessing (e.g., encoding) and for interpreting the predictive model’s outputs.

#### ***Fuel Type (cod\_cbr).***

The `cod_cbr` variable describes the primary fuel or fuel combination used by the vehicle. It contains 11 unique categories: {ES, GO, ES/GP, GP/ES, EH, GH, ES/GN, GN/ES, FE, GN, GL}. Here, ES corresponds to petrol (essence), GO to diesel (gazole), FE to low-emission fuels (faible émission), while other codes represent hybrid configurations or alternative fuels such as natural gas (GN) or liquefied gas (GL). Hybrids are denoted with slashes indicating the combination of energy sources (e.g., ES/GP).

#### ***Hybrid Indicator (hybride).***

This binary variable indicates whether a vehicle is a hybrid, taking values `non` or `oui`. While partially redundant with some `cod_cbr` entries (e.g., hybrid fuel codes), it can serve as a simplified indicator in certain analyses.

#### ***Transmission (typ\_boite\_nb\_rapp).***

The original transmission variable combines gearbox type and number of gears, with 14 unique values (e.g., M 6, A 7, D 5). To improve interpretability, we split it into two separate variables:

- `gearbox_type`: 5 categories (M = manual, A = automatic, D = dual-clutch, V = continuously variable transmission, S = semi-automatic).
- `num_gears`: numeric variable representing the number of forward gears, with 7 distinct values: {0, 4, 5, 6, 7, 8, 9}.

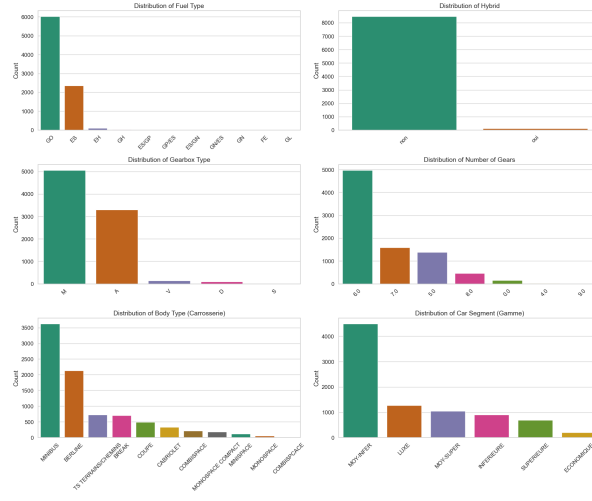
#### ***Body Style (Carrosserie).***

This feature identifies the vehicle’s body configuration, with 11 categories including common types (BERLINE, BREAK, COUPE, CABRIOLET) and specialized designs (TS TERRAINS/CHEMINS for off-road vehicles, COMBISPACE, MINISPACE, MONOSPACE, MINIBUS).

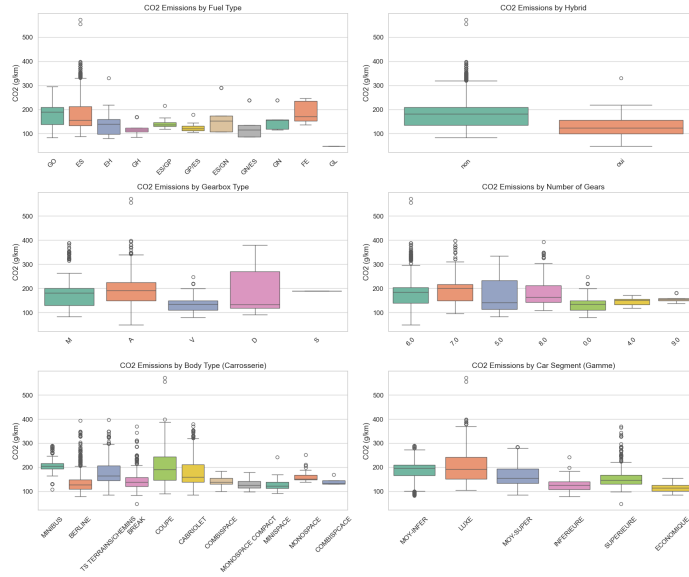
#### ***Market Segment (gamme).***

This categorical variable groups vehicles into six market segments: {ECONOMIQUE, INFERIEURE, MOY-INFER, MOY-SUPER, SUPERIEURE, LUXE}. These segments reflect pricing and positioning differences that may indirectly influence CO<sub>2</sub> emissions through associated design choices and performance levels.

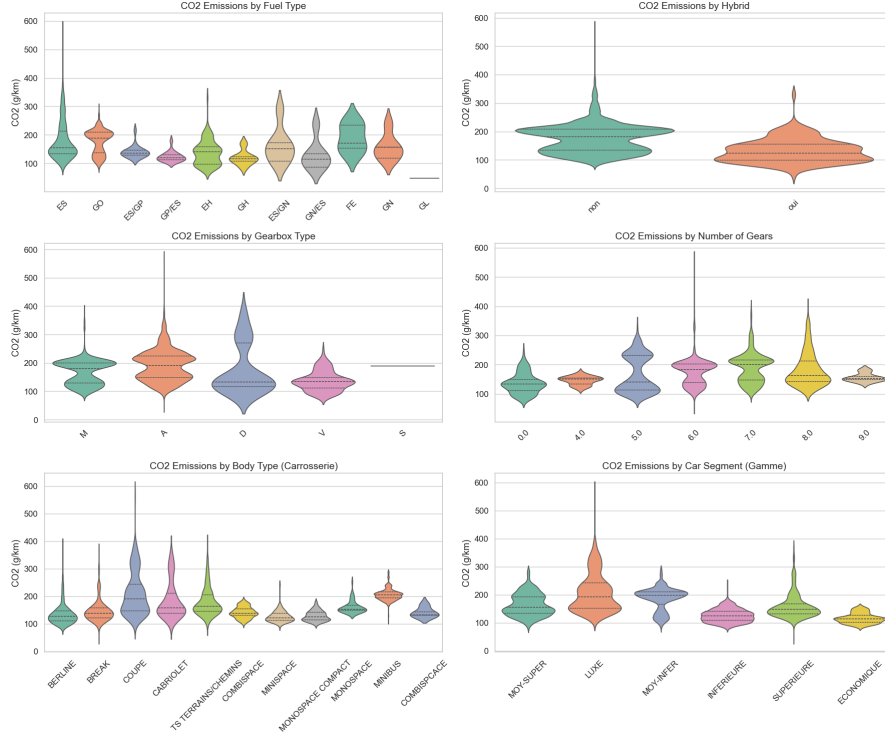
Overall, these categorical variables provide rich qualitative information on vehicle technology, configuration, and market positioning, and will be incorporated into the modeling phase through appropriate encoding strategies.



**Fig. 6:** Count plots displaying the distribution of categorical features.



**Fig. 7:** Boxplots displaying the relationships between categorical features and CO<sub>2</sub> emissions.



**Fig. 8:** Violin plots showing the distribution of CO<sub>2</sub> emissions across categorical features.

We first examine the distribution of categorical vehicle features (Figure 6), where the count plots reveal notable imbalances across categories. Fuel type (`cod_cbr`) is heavily skewed toward conventional fuels (GO, ES), while low-emission fuels (FE, GL) and hybrid combinations are relatively rare. Similarly, the hybrid indicator (`hybride`) is strongly imbalanced, with the vast majority of vehicles being non-hybrids. Gearbox type (`gearbox_type`) is dominated by manual and automatic transmissions, and the number of gears (`num_gears`) is concentrated around 6 and 7, reflecting typical modern configurations. Body type (`Carrosserie`) is primarily minibus or berline, and car segment (`gamme`) is skewed toward mid- and lower-end categories (MOY-INFER, MOY-SUPER, LUXE), with sporty, niche, or premium vehicles being less frequent.

To investigate how these categorical imbalances relate to vehicle CO<sub>2</sub> emissions, we analyze both the box and violin plots (Figures 7 and 8). The box plots illustrate differences in central tendency and spread for CO<sub>2</sub> emissions across categories, while the violin plots provide additional detail about the underlying distribution, density, and presence of outliers. Low-emission fuels consistently correspond to lower median CO<sub>2</sub>, whereas conventional fuels display higher medians and substantial variability, including extreme outliers. Hybrid vehicles, though infrequent, show notably lower

and more tightly clustered CO<sub>2</sub> emissions (around 120g/km) compared to non-hybrids (180–200g/km), underscoring the effectiveness of hybrid technology.

Gearbox type and number of gears show a more nuanced influence. Across commonly observed categories, median emissions are similar, but certain groups—particularly diesels and vehicles with higher gear counts—are characterized by wider spreads and more pronounced outliers (Figure 8), suggesting interactions with fuel type and engine characteristics. Body type and car segment also demonstrate clear differences: sporty, premium, or luxury vehicles generally emit more CO<sub>2</sub>, whereas economy and compact models show lower values.

Taken together, the combined evidence from count, box, and violin plots indicates that while models may be dominated by prevalent category groups, features such as fuel type, hybrid status, and car segment exhibit the strongest and most consistent associations with CO<sub>2</sub> emissions. Gearbox type, number of gears, and body type contribute additional explanatory power through their influence on variability and the presence of outliers.

## 4 Machine Learning Models

In this section, we present the machine learning models applied to predict CO<sub>2</sub> emissions of vehicles. For each model, we provide a brief description, the performance results, and the feature importance. Model evaluation relies on several complementary performance metrics. The cross-validated  $R^2$  (CV  $R^2$ ) is used to assess the model’s generalization ability across different data splits, mitigating the risk of overfitting. The test  $R^2$  measures the proportion of variance in CO<sub>2</sub> emissions explained by the model on unseen data, providing an interpretable indication of predictive accuracy. Finally, the test Root Mean Squared Error (RMSE) quantifies the average magnitude of prediction errors in the same unit as the target variable, making it directly comparable to the observed emissions. Together, these metrics offer a balanced evaluation of both explanatory power and predictive reliability.

### 4.1 Lasso Regression

Lasso (Least Absolute Shrinkage and Selection Operator) regression is a linear model that performs both regularization and feature selection by penalizing the absolute values of the regression coefficients (Tibshirani, 1996). This penalty encourages sparsity in the model, effectively setting less important feature coefficients to zero, which simplifies the model and can improve interpretability. Lasso is particularly useful when there are many correlated features, as it can automatically select the most relevant predictors while reducing overfitting.

#### 4.1.1 Results

#### 4.1.2 Feature Importance

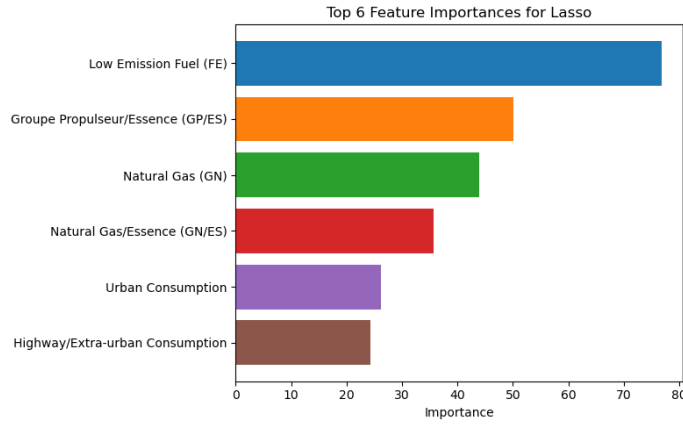
Top features (importance in parentheses):

- Low Emission Fuel (FE) (76.79)

| Metric          | Value                  |
|-----------------|------------------------|
| Best Parameters | {'model__alpha': 0.01} |
| Best CV $R^2$   | 0.9964                 |
| Test $R^2$      | 0.9977                 |
| Test RMSE       | 2.4213                 |

**Table 2:** Performance metrics for Lasso regression.

- Groupe Propulseur/Essence (GP/ES) (50.08)
- Natural Gas (GN) (43.98)
- Natural Gas/Essence (GN/ES) (35.73)
- Urban Consumption (26.13)
- Highway/Extra-urban Consumption (24.26)



**Fig. 9:** Feature importance according to the Lasso model.

The high Test  $R^2$  (0.9977) and low RMSE indicate that Lasso achieves excellent predictive accuracy on unseen data, confirming that a linear combination of a few key features explains most of the variance in CO<sub>2</sub> emissions. The sparsity enforced by Lasso highlights that fuel type dominates emissions, while consumption variables have moderate effects. This suggests that linear effects of fuel type are strong, but some non-linear relationships may remain unmodeled.

## 4.2 Ridge Regression

Ridge regression is a linear model that applies  $L_2$  regularization to shrink coefficients toward zero but does not enforce sparsity (Hoerl and Kennard, 1970). This helps to reduce the impact of multicollinearity among features and prevent overfitting, especially when the number of features is large. Ridge regression is useful when most features are expected to contribute to the prediction but with varying magnitudes.

#### 4.2.1 Results

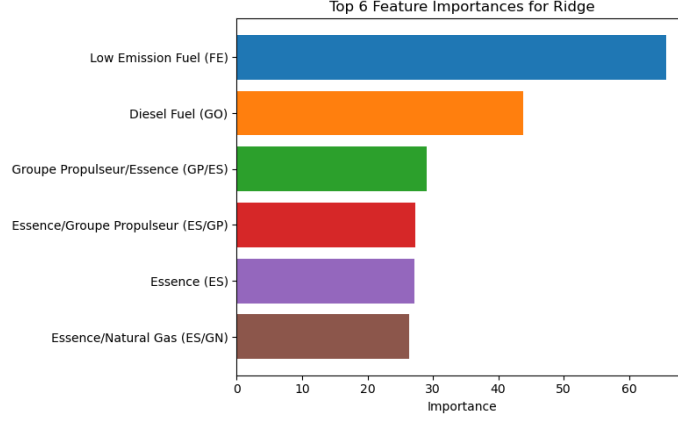
| Metric          | Value                            |
|-----------------|----------------------------------|
| Best Parameters | $\{\text{'model\_alpha': 0.1}\}$ |
| Best CV $R^2$   | 0.9968                           |
| Test $R^2$      | 0.9974                           |
| Test RMSE       | 2.5943                           |

**Table 3:** Performance metrics for Ridge regression.

#### 4.2.2 Feature Importance

Top features (importance in parentheses):

- Low Emission Fuel (FE) (65.61)
- Diesel Fuel (GO) (43.84)
- Groupe Propulseur/Essence (GP/ES) (29.04)
- Essence/Groupe Propulseur (ES/GP) (27.31)
- Essence (ES) (27.20)
- Essence/Natural Gas (ES/GN) (26.38)



**Fig. 10:** Feature importance according to the Ridge model.

Ridge achieves a high Test  $R^2$  (0.9974) and low RMSE, similar to Lasso, confirming strong predictive performance. Unlike Lasso, Ridge spreads importance across multiple fuel types, suggesting CO<sub>2</sub> emissions are influenced by combinations of fuels rather than a single category. The performance metrics indicate that the linear approximation is sufficient for most variance, though Ridge captures more subtle, cumulative effects across correlated fuel features.



### 4.3 Random Forest

Random Forest is an ensemble learning method that builds multiple decision trees on different bootstrap samples of the data and averages their predictions (Breiman, 2001). This approach reduces overfitting and improves generalization compared to a single decision tree. Random Forest can handle non-linear relationships and interactions between features, and it provides feature importance metrics based on how much each feature reduces impurity across all trees. It is robust to noisy features and widely used in regression and classification tasks.

#### 4.3.1 Results

| Metric          | Value                                                 |
|-----------------|-------------------------------------------------------|
| Best Parameters | {'model__max_depth': None, 'model__n_estimators': 50} |
| Best CV $R^2$   | 0.9959                                                |
| Test $R^2$      | 0.9986                                                |
| Test RMSE       | 1.8752                                                |

**Table 4:** Performance metrics for Random Forest.

#### 4.3.2 Feature Importance

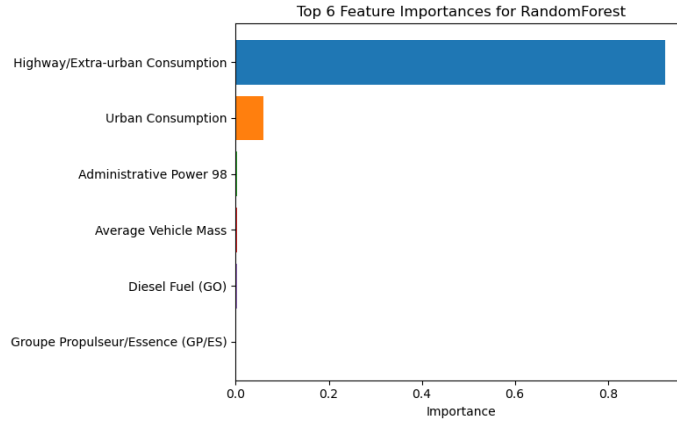
Top features (importance in parentheses):

- Highway/Extra-urban Consumption (0.92)
- Urban Consumption (0.06)
- Administrative Power 98 (0.00)
- Average Vehicle Mass (0.00)
- Diesel Fuel (GO) (0.00)
- Groupe Propulseur/Essence (GP/ES) (0.00)

Random Forest achieves the highest Test  $R^2$  (0.9986) and lowest RMSE (1.8752), showing superior predictive power compared to linear models. Feature importance indicates that vehicle efficiency and driving patterns outweigh fuel type, reflecting non-linear effects and interactions that linear models cannot capture. The combination of high performance and consumption-driven importance suggests engineering and usage factors strongly determine real-world emissions, offering actionable insight beyond fuel selection.

### 4.4 XGBoost

XGBoost (Extreme Gradient Boosting) is a gradient boosting algorithm that builds an ensemble of decision trees sequentially (Chen and Guestrin, 2016). Each tree is trained to correct the errors of the previous ones by optimizing a differentiable loss function. XGBoost includes regularization terms to prevent overfitting, and it is highly



**Fig. 11:** Feature importance according to the Random Forest model.

efficient and scalable. It handles non-linearities and interactions well and often achieves state-of-the-art performance on tabular datasets.

#### 4.4.1 Results

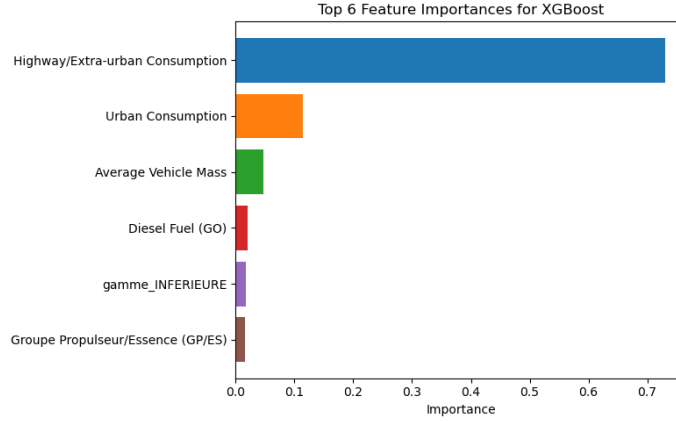
| Metric          | Value                                                                           |
|-----------------|---------------------------------------------------------------------------------|
| Best Parameters | {'model__learning_rate': 0.1, 'model__max_depth': 3, 'model__n_estimators': 50} |
| Best CV $R^2$   | 0.9912                                                                          |
| Test $R^2$      | 0.9937                                                                          |
| Test RMSE       | 4.0523                                                                          |

**Table 5:** Performance metrics for XGBoost.

#### 4.4.2 Feature Importance

Top features (importance in parentheses):

- Highway/Extra-urban Consumption (0.73)
- Urban Consumption (0.11)
- Average Vehicle Mass (0.05)
- Diesel Fuel (GO) (0.02)
- gamme\_INFERIEURE (0.02)
- Groupe Propulseur/Essence (GP/ES) (0.02)



**Fig. 12:** Feature importance according to the XGBoost model.

XGBoost shows a slightly lower Test  $R^2$  (0.9937) and higher RMSE (4.05) compared to Random Forest, indicating that for this dataset, simpler tree ensembles generalize slightly better. The dominance of consumption features suggests emissions are largely driven by non-linear interactions of vehicle usage and efficiency. Fuel type contributes marginally, reinforcing that policy measures should also consider vehicle performance and driving behavior.

## 5 Comparison Between Manufacturer Advertising in Auto Plus and Model-Derived Feature Importance

### 5.1 Introduction to Auto Plus and 2014 Advertising Context

*Auto Plus* is a widely read French automotive magazine, known for publishing vehicle rankings, technical tests, and consumer-oriented buying advice (Pennec, 2014). Among its recurring features is the “Classement des véhicules propres” (Clean Vehicle Ranking), which highlights cars with the lowest environmental impact.

In its 2014 “Palmarès” issue, *Auto Plus* identified hybridization as a notable technical differentiator among manufacturers, alongside stylistic and economic strategies: “Une année de plus à oublier pour bien des constructeurs. . . ça va bien pour ceux qui font preuve d’audace stylistique (SUV), économique (low cost), technique (hybridation)” (Pennec, 2014). This emphasis reflects broader industry trends in France at the time, where hybrid systems were frequently promoted as “0 g/km” emission solutions (tailpipe only), often without reference to upstream emissions (ADEME, 2014).

Beyond hybridization, manufacturers actively marketed other fuel-saving and emissions-reducing technologies, such as:

- **Engine downsizing with turbocharging**, promoted for combining high performance with reduced fuel consumption (ACEA, 2014).
- **Start/Stop systems**, advertised as cutting idle fuel use in urban driving conditions (ADEME, 2014).

- **Efficient transmissions** (dual-clutch gearboxes, higher gear counts) and low rolling resistance tires, presented as incremental efficiency boosters (ACEA, 2014).

These marketing narratives tended to focus on visible technological features rather than directly reporting the quantitative, measurable performance metrics—such as certified urban and extra-urban fuel consumption—that most strongly influence actual CO<sub>2</sub> emissions.

## 5.2 Model-Derived Feature Importance (Summary of Results)

Our machine learning models, trained on real vehicle data, identified the following as the most important predictors of CO<sub>2</sub> emissions:

| Model         | Best CV $R^2$ | Test $R^2$ | Test RMSE | Top Features (Importance Scores)                                                                |
|---------------|---------------|------------|-----------|-------------------------------------------------------------------------------------------------|
| Lasso         | 0.9964        | 0.9977     | 2.4213    | Fuel types (FE 76.8, GP/ES 50.1, GN 44.0, GN/ES 35.7), Urban (26.1), Extra-urban (24.3)         |
| Ridge         | 0.9968        | 0.9974     | 2.5943    | FE 65.6, GO 43.8, GP/ES 29.0, ES/GP 27.3, ES 27.2, ES/GN 26.4                                   |
| Random Forest | 0.9959        | 0.9986     | 1.8752    | Extra-urban (0.92), Urban (0.06), Power, Mass, GO, GP/ES (near 0)                               |
| XGBoost       | 0.9912        | 0.9937     | 4.0523    | Extra-urban (0.73), Urban (0.11), Mass (0.05), GO (0.02), gamme_INFERIEURE (0.02), GP/ES (0.02) |

**Table 6:** Summary of model performance and top features for predicting CO<sub>2</sub> emissions.

## 5.3 Comparison

The analysis reveals a clear divergence between the focus of advertising in *Auto Plus* and the findings from our machine learning models. In 2014, manufacturers promoted visible technologies such as hybrid systems, start/stop mechanisms, and turbocharged small engines, framing them as major innovations for reducing emissions. However, our models indicate that the most influential factors on CO<sub>2</sub> output are urban and extra-urban fuel consumption and fuel type—variables directly tied to measurable emissions. While mass and power also emerge as relevant predictors in tree-based models, they were not commonly highlighted in ecological marketing. Furthermore, features like start/stop systems or special gearboxes may influence emissions indirectly through reduced fuel consumption but are not primary standalone predictors in the dataset. This contrast underscores how marketing narratives often prioritize technological branding over scientifically measurable performance, suggesting that future eco-performance communication could benefit from a stronger alignment with data-driven evidence.

## 6 Web Application Architecture and Functionality

The developed web application is an interactive platform for predicting vehicle CO<sub>2</sub> emissions and determining the corresponding *bonus-malus* status under the French

2014 policy. It is implemented using the **Dash** framework (built on top of Plotly and Flask) and styled with Bootstrap components for a responsive and user-friendly interface.

## 6.1 Data and Model Loading

At initialization, the application loads:

- **Preprocessing pipeline:** a **scikit-learn** transformer (`preprocessor.pkl`) that applies feature transformations such as encoding and scaling.
- **Trained prediction models:** four algorithms—Lasso, Ridge, Random Forest, and XGBoost—each stored in serialized form (`.pkl` or `.json`).
- **Model performance summary:** a CSV file (`model_results.csv`) containing cross-validation and test metrics, later displayed to the user.
- **Bonus–malus policy data:** a CSV file (`bonus_malus_france_2014_combined.csv`) defining CO<sub>2</sub> thresholds and corresponding tax/bonus amounts.

The application distinguishes **numeric** features (e.g., mass, power, fuel consumption) from **categorical** features (e.g., fuel type, hybrid indicator, gearbox type), ensuring consistent preprocessing.

## 6.2 Application Layout

The user interface is structured into several components:

1. **Model Selection:** dropdown menu to choose the prediction algorithm.
2. **Vehicle Specifications:**
  - Numeric inputs for administrative power, urban and extra-urban fuel consumption, and mass.
  - Dropdowns for number of gears, fuel type, hybrid status, gearbox type, body type, and vehicle range (*gamme*).
  - All categorical values are presented with human-readable labels (e.g., "ES" → "Petrol (ES)", "oui" → "Yes").
3. **Action Button:** triggers the prediction process.
4. **Results Display:**
  - An alert summarizing predicted CO<sub>2</sub> emissions and applicable bonus/malus outcome.
  - A table showing the interpreted user inputs (with descriptive labels).
  - A table summarizing each model's performance metrics.

## 6.3 Prediction Pipeline

Upon user action, the following sequence occurs:

1. **Input Collection and Validation:** numeric inputs are coerced into numeric types; categorical inputs are cast as strings; features are ordered for the models.

2. **Preprocessing:** the preprocessor transforms inputs into a model-ready feature matrix (scaling, encoding, etc.).
3. **Model Inference:** the chosen model estimates CO<sub>2</sub> emissions (g/km).
4. **Bonus–Malus Evaluation:** predicted CO<sub>2</sub> is compared to 2014 thresholds and classified into *bonus*, *neutral*, or *malus*.
5. **Result Formatting:** feature names are mapped to descriptive labels, categorical codes expanded, and outputs displayed in Bootstrap-styled tables.

## 6.4 User Experience Enhancements

The application includes:

- Responsive design via Bootstrap for desktop and mobile devices.
- Transparency via a model performance summary for algorithm comparison.
- Error handling: prediction errors are caught and shown to the user as alerts.

## 6.5 Deployment

The application runs on a Dash `app` instance, served through a Flask backend (`server = app.server`), enabling both local execution and cloud deployment (e.g., Heroku, Render, or Dockerized environments).

## 7 Conclusion

This project successfully developed an interactive web application for predicting vehicle CO<sub>2</sub> emissions and determining their classification under the French 2014 *bonus–malus* scheme. By integrating multiple machine learning models (Lasso, Ridge, Random Forest, and XGBoost) with a unified preprocessing pipeline, the application delivers accurate predictions while maintaining transparency through the display of model performance metrics. The use of the Dash framework, combined with Bootstrap components, enabled the creation of a responsive and user-friendly interface, making the system accessible for both technical and non-technical audiences.

In addition to predictive functionality, the tool enhances user engagement by translating technical inputs into human-readable labels, applying semantic visual cues for results, and clearly explaining the impact of emissions on policy incentives. This approach ensures that the tool is not only technically robust but also practical and informative for decision-making in the context of environmental policy.

While the current version meets its design objectives, several improvements could enhance its performance and scope in future iterations:

- **Real-time data integration:** connect to live vehicle databases or APIs to automatically retrieve up-to-date specifications.
- **Expanded policy coverage:** include bonus–malus schemes from other years or countries to broaden applicability.
- **User account features:** enable users to save, compare, and export predictions.
- **Mobile optimisation:** further refine the interface for improved usability on smaller devices.

- **Model update automation:** implement scheduled retraining with new data to maintain accuracy.
- **Multilingual support:** offer translations for all interface elements to reach a wider audience.

Overall, the application demonstrates how predictive modelling and interactive visualisation can be effectively combined to inform environmental decision-making, while offering a strong foundation for future expansion and refinement.

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